

Product, Quotient, and Trigonometric Derivatives

ANSWER KEY



The product rule

- 1 Differentiate $f(x) = (3x^2 - 1)(x^3 + 2x)$ using the product rule.
 $f'(x) = 6x(x^3 + 2x) + (3x^2 - 1)(3x^2 + 2) = 6x^4 + 12x^2 + 9x^4 + 6x^2 - 3x^2 - 2 = 15x^4 + 15x^2 - 2.$
- 2 Differentiate $f(x) = x^2 \sin x$.
 $f'(x) = 2x \sin x + x^2 \cos x.$
- 3 Differentiate $f(x) = e^x \cos x$.
 $f'(x) = e^x \cos x - e^x \sin x = e^x (\cos x - \sin x).$

The quotient rule

- 4 Differentiate $f(x) = \frac{3x-1}{x+4}$.
 $f'(x) = \frac{3(x+4) - (3x-1)(1)}{(x+4)^2} = \frac{3x+12-3x+1}{(x+4)^2} = \frac{13}{(x+4)^2}.$
- 5 Differentiate $f(x) = \frac{x^2}{x^2+1}$.
 $f'(x) = \frac{2x(x^2+1) - x^2(2x)}{(x^2+1)^2} = \frac{2x^3+2x-2x^3}{(x^2+1)^2} = \frac{2x}{(x^2+1)^2}.$
- 6 Differentiate $f(x) = \frac{\sin x}{x}$.
 $f'(x) = \frac{x \cos x - \sin x}{x^2}.$

Derivatives of the six trig functions

- 7 State the derivative of each trig function:

a $\frac{d}{dx} \sin x = \cos x$	b $\frac{d}{dx} \cos x = -\sin x$	c $\frac{d}{dx} \tan x = \sec^2 x$
d $\frac{d}{dx} \cot x = -\csc^2 x$	e $\frac{d}{dx} \sec x = \sec x \tan x$	f $\frac{d}{dx} \csc x = -\csc x \cot x$
- 8 Differentiate $f(x) = x \tan x$.
 $f'(x) = \tan x + x \sec^2 x.$
- 9 Differentiate $f(x) = \sec x \cdot \csc x$.
 $f'(x) = \sec x \tan x \csc x + \sec x (-\csc x \cot x) = \sec x \csc x (\tan x - \cot x).$

Combining product, quotient, and trig rules

10 Differentiate $f(x) = \frac{e^x}{\sin x}$.

$$f'(x) = \frac{e^x \sin x - e^x \cos x}{\sin^2 x} = \frac{e^x(\sin x - \cos x)}{\sin^2 x}.$$

- 11 Find the equation of the tangent line to $f(x) = x \cos x$ at $x = 0$.
 $f(0) = 0$. $f'(x) = \cos x - x \sin x$; $f'(0) = 1$. Tangent: $y - 0 = 1(x - 0)$, i.e. $y = x$.

Derivatives using a table of values

- 12 Given $f(3) = 2$, $f'(3) = -1$, $g(3) = 4$, $g'(3) = 5$, find $h'(3)$ if $h(x) = f(x)g(x)$.
 $h'(x) = f'(x)g(x) + f(x)g'(x)$; $h'(3) = (-1)(4) + (2)(5) = -4 + 10 = 6$.

- 13 Using the same table, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.
 $k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$; $k'(3) = \frac{(-1)(4) - (2)(5)}{4^2} = \frac{-4 - 10}{16} = -\frac{14}{16} = -\frac{7}{8}$.

Second derivatives with product and quotient rules

- 14 Find $f''(x)$ for $f(x) = x \sin x$.
 $f'(x) = \sin x + x \cos x$. $f''(x) = \cos x + \cos x - x \sin x = 2 \cos x - x \sin x$.